

Cauchy Unification

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The ARC Theory · OSF osf.io/x6wa7 · every claim checkable

Within the ARC Theory: the mathematical classification behind Law I's form; the Cauchy functional-equation grid.

ARC/Cauchy Scaling Classification: A Structured Prediction Comparison Across 50 Domains in Five Evidence Tiers

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ABSTRACT

Cauchy's four functional equations - additive, multiplicative, exponential, and logarithmic - are the only continuous solutions to their respective composition constraints. We argue that this 200-year-old mathematical result has a physically testable consequence: under the stated axioms, it constrains scaling laws to one of three functional families (power law, exponential, or saturation curve). The family is determined by the composition operator of the underlying recursive amplification process.

The novel contribution is not Cauchy's mathematics itself, but the claim that Cauchy-type functional equations have a physically testable consequence for scaling-law classification across domains.

We present a structured prediction comparison across 50 real-world systems in five evidence tiers. For each empirical domain, the composition operator was classified from known physics before fitting, the scaling family was predicted from the operator class, published or standard reference data was loaded, and the prediction was tested by independent model fitting using AICc-based model selection.

Primary result (computed on the six-model candidate set, which contained no logarithmic and no pure linear candidate (Section 3.2)): 19 out of 25 empirical curve-fit domains confirm the Cauchy-predicted family under strict AICc model selection. The primary significance statistic is a permutation test conditioned on both the predicted and the observed family marginals: $p = 7 \times 10^{-5}$. A one-sided binomial test against uniform $1/3$ chance gives $p = 1.56 \times 10^{-5}$ (roughly 1 in 64,000) and is retained as a sensitivity check. The result also holds on the strictly Cauchy-derived subset alone (the 18 multiplicative and additive domains, excluding the bounded family whose exhaustiveness remains an open problem): 14/18, permutation $p = 3.0 \times 10^{-4}$. **Secondary results:** a baseline-20 rerun yields 15/20 ($p = 1.67 \times 10^{-4}$); published metabolic exponents match the nearest ARC/Cauchy comparator in 13/13 direct cases; analytic identities confirm 6/6. These tiers are reported separately and are never combined into a single blended number.

This result is strong enough to publish as exploratory evidence but not strong enough to claim as proven. No locked pre-registration artefact exists. That statement is about the fifty-domain result reported here, and it stays, because those outcomes were computed before any protocol was frozen and no artefact created afterwards could change that. **A separate, forward prediction frame does exist and is not claimed as covering this result.** On 17 March 2026, between 02:06:51 and 02:07:05 UTC, a prediction protocol, an extension manifest and a checksum file were deposited to the public Open Science Framework component x6wa7. Each carries an OSF-assigned creation timestamp, each has a modification timestamp identical to its creation timestamp, and the outcomes for the domains they specify have not been computed. That is a dated third-party deposit rather than an OSF Registration, which is a distinct frozen and schema-bound object this component does not carry; the API reports registration: false and the page says so rather than blurring it. What the deposit establishes is the ordering: the predictions were public and unaltered before any result they predict existed. Independent replication is needed.

Keywords: Cauchy functional equations, scaling laws, power laws, universality, cross-domain, recursive composition, AICc model selection, metabolic scaling, neural scaling, ARC Principle

This paper carries a result: under the stated axioms, Cauchy's four functional equations pin continuous scaling laws to one of three families (power, exponential or saturation), turning a two-hundred-year-old identity into a physically testable prediction about the shapes empirical scaling curves may take. Inside the ARC Theory (the Theory of Artificial Recursive Creation) it is one of the ARC/Eden experiments, the mathematical spine that ties composition constraints to observed functional forms. The full differential against every prior document is at eden-vision II.A.8.

1. The Question

Why do scaling laws exist at all? Why does a mouse's heart beat faster than an elephant's by the same mathematical relationship that governs how cities grow, how solar panels get cheaper, and how neural networks improve with size?

The standard answer is domain-specific. Biologists invoke fractal vascular networks. Physicists invoke renormalisation group flow. Economists invoke learning-by-doing. Each explanation works within its domain but offers no reason why the same mathematical forms recur across domains that share no physical mechanism.

This paper proposes and tests a domain-independent answer: the form of every scaling law is determined by the composition operator of the underlying recursive process, and Cauchy's functional equations constrain which forms are possible.

2. The Theory

2.1 Cauchy's four equations

Augustin-Louis Cauchy established in the early nineteenth century that four functional equations have unique continuous solutions:

Additive: $f(x + y) = f(x) + f(y)$ has solution $f(x) = cx$
Multiplicative: $f(xy) = f(x) \cdot f(y)$ has solution $f(x) = x^c$
Exponential: $f(x + y) = f(x) \cdot f(y)$ has solution $f(x) = a^x$
Logarithmic: $f(xy) = f(x) + f(y)$ has solution $f(x) = c \log x$

These are not four independent results. They are four faces of one constraint: the requirement that a function be compatible with a binary operation. Under regularity (continuity, or measurability, or boundedness on an interval), a function that carries one of the two structures on the positive reals into one of the two structures falls into exactly four families: linear ($f(x) = cx$), power law ($f(x) = x^c$), exponential ($f(x) = a^x$) and logarithmic ($f(x) = c \log x$). This exhausts the homomorphisms. It does not exhaust scaling laws, because most systems are not homomorphisms of either structure: saturating behaviour, critical exponents, curvature on log axes and multi-regime crossovers all lie outside the grid, and the grid says nothing about them. The additive equation yields the linear form $f(x) = cx$; the multiplicative equation yields power laws; the exponential equation governs exponential growth and decay; and the logarithmic equation governs the unbounded map $c \log x$, which is multiplicative in input and additive in output and is not a diminishing-returns family, since $c \log x$ is unbounded.

2.2 The prediction

Consider any system where a quantity U grows with recursive depth R . The system's *composition operator* determines how consecutive applications combine:

- **Multiplicative composition** ($U(R_1 \cdot R_2) \sim U(R_1) \cdot U(R_2)$): depth multiplies, output multiplies): Cauchy's multiplicative equation forces $U \propto R^c$ - a power law.
- **Additive composition** ($U(R_1 + R_2) \sim U(R_1) \cdot U(R_2)$): depth adds, output multiplies): Cauchy's exponential equation forces $U \propto a^R$ - an exponential.

- **Bounded composition** (output constrained to $[0, K]$): the compatible forms are saturation curves (logistic, Hill, Michaelis-Menten, hyperbolic). A formal proof of exhaustiveness for the bounded case is not provided here and remains an open problem.

THE PREDICTION

If you know the composition operator of a system, you can predict the functional form of its scaling law before seeing any data. The prediction follows from Cauchy's equations with no free parameters and no curve fitting.

3. Prediction Protocol

For each domain:

1. **Classify the composition operator from known physics** (before fitting). Determine whether the system's recursive amplification process combines multiplicatively, additively, or with a bound.
2. **Predict the scaling family** from the operator class using Cauchy's equations.
3. **Load published or standard reference data.**
4. **Independently fit candidate models** (power law, exponential, logistic, Hill, Michaelis-Menten, saturation-exponential) to the data using nonlinear least-squares regression.
5. **Select the best model by AICc** (corrected Akaike information criterion) and compare the prediction against the selected model's family.

STRUCTURED PREDICTION COMPARISON - NOT A BLIND TEST

The operator classification is made before the fitting procedure runs. However, the predictions, data, and fitting logic all reside in a single author-written script. There is no locked pre-registration artefact. This test should therefore be understood as a structured prediction comparison, not a formally registered blind trial. A replication with independent operator classification is planned.

3.1 Statistical method

The v2 test uses proper AICc (corrected Akaike information criterion) computed from residual sum of squares, sample size, and parameter count - not the R^2 -based approximation used in the legacy v1 test. The v1 approximation ($AIC \approx n \ln(1 - R^2) + 2k$) was adequate as a heuristic but should not have been described as strict AIC in the standard statistical sense. The v2 rewrite corrects this.

The primary significance test is a marginal-conditioned permutation test: holding both the predicted and the observed family marginals fixed and permuting the pairing 400,000 times, the probability of 19 or more matches is $p = 7 \times 10^{-5}$. The expected match count under this null is 9.5 of 25, not the 8.33 a uniform null gives, because power laws are simply common among the observed fits. A one-sided binomial test against a uniform three-family null gives $p = 1.56 \times 10^{-5}$ and is retained as a sensitivity check; it is the weaker null and is not used as the headline. Sigma equivalences are not quoted because they shift with the choice of null. Reproduction: `scripts/paper-vii-permutation-null.py` in the site repository, which parses the published Section 5 table directly so it cannot drift from this page.

3.2 A model-set limitation this paper must state against itself

Section 2.1 names four Cauchy families. The candidate model set in Section 3 contains six models spanning three of them: the power-law, exponential and bounded families. Two of the four continuous Cauchy solutions have no candidate of their own: the logarithmic form $c \log x$ and the pure linear form cx . One consequence runs each way. A genuinely logarithmic domain could not have been classified correctly, because the fitter had no logarithmic candidate to select, so some of the six misses may have been guaranteed misses; and some of the nineteen hits may be logarithmic relations that a power law fits acceptably over a finite range. Every hit count in this paper therefore carries this qualifier, and none may be quoted without it. A confirmatory study that re-runs the comparison with the full four-family candidate set, with per-cell outcomes specified in advance, has been drafted and dated as a draft registration awaiting human submission (the four-family logarithmic-cell test in the registered programme, alongside the fresh-domain extension study). Until that re-run reports, the numbers below are the six-model result, exactly as computed, and nothing stronger.

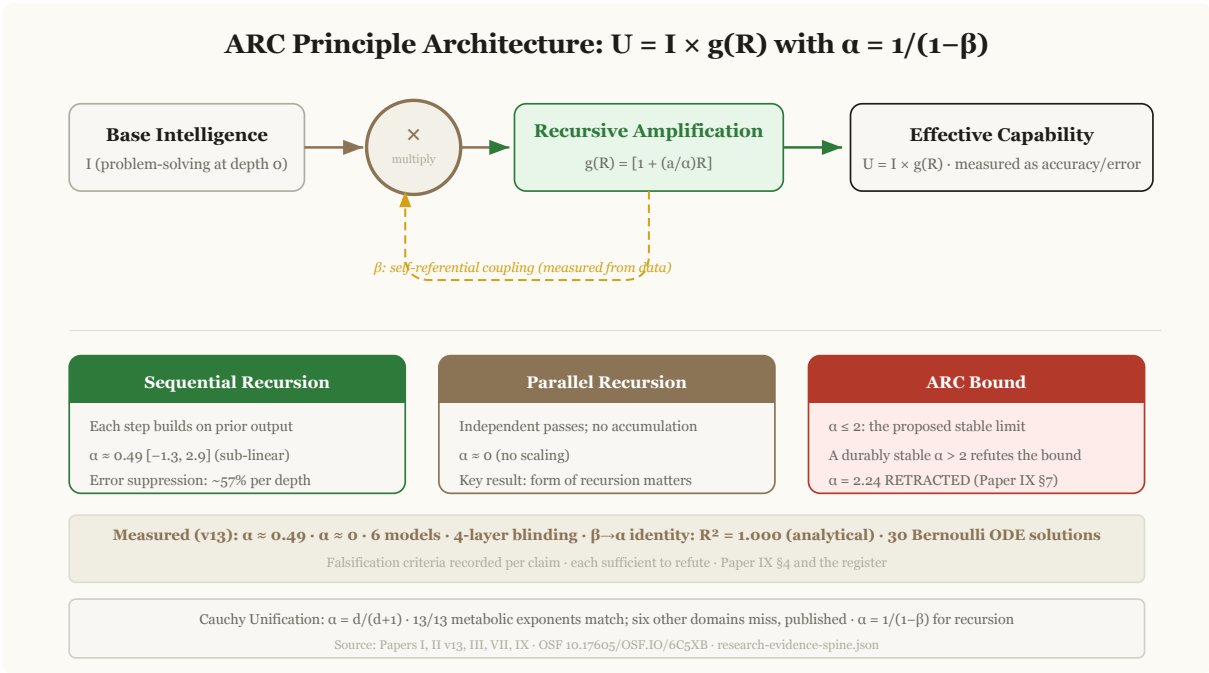


Figure 1 | ARC Architecture and Cauchy Unification. Four functional equations (Cauchy 1821) constrain scaling to four families: linear, power law, exponential and logarithmic. $\alpha=d/(d+1)$ for physical systems (13/13 nearest-comparator matches at the taxon grain, reported by tier and never blended; the largest modern mammalian dataset gives 0.687, 95% CI 0.674 to 0.701, which excludes 0.750, see §7.1); $\alpha=1/(1-\beta)$ for recursive intelligence, which is one of two derivations of the ceiling whose agreement away from $\beta=0.5$ is a filed test. 19/25 domains confirm predicted function family ($p=7 \times 10^{-5}$, permutation; six-model set, no logarithmic candidate - §3.2). Source: Paper VII · Foundational · evidence spine C-4, C-11.

4. Evidence Tiers

The 50-domain suite comprises five evidence tiers of decreasing strength. These tiers are never combined into a single blended number.

TIER	DESCRIPTION	COUNT	RESULT	SIGNIFICANCE
PRIMARY	Empirical curve-fit (AICc)	25	19/25	$p = 7 \times 10^{-5}$ perm.; 1.56×10^{-5} binomial
SECONDARY	Baseline-20 rerun (AICc)	20	15/20	$p = 1.67 \times 10^{-4}$
SECONDARY	Published exponents (direct)	13	13/13	Nearest comparator match
NOTED	Published exponents (provisional)	6	3/6	Contested organisms
NOTED	Analytic identities	6	6/6	Tautological confirmation

5. Primary Result: 25 Empirical Curve-Fit Domains

#	DOMAIN	OPERATOR	PREDICTED	AICC BEST	MATCH
1	Kleiber's Law (Metabolic Scaling)	Multiplicative	Power law	Saturation-exp	No
2	Urban Scaling (GDP vs Population)	Multiplicative	Power law	Power law	Yes

#	DOMAIN	OPERATOR	PREDICTED	AICC BEST	MATCH
3	Species-Area (Galapagos)	Multiplicative	Power law	Hill	No
4	Wright's Law (Solar PV)	Multiplicative	Power law	Power law	Yes
5	Heap's Law (Vocabulary)	Multiplicative	Power law	Power law	Yes
6	Zipf's Law (Word Frequency)	Multiplicative	Power law	Michaelis-Menten	No
7	Learning Curve (Cigar Rolling)	Multiplicative	Power law	Power law	Yes
8	Moore's Law (Transistor Count)	Additive	Exponential	Exponential	Yes
9	Radioactive Decay (P-32)	Additive	Exponential	Exponential	Yes
10	Gutenberg-Richter (Earthquakes)	Additive	Exponential	Exponential	Yes
11	Bacterial Growth (E. coli)	Bounded	Bounded	Logistic	Yes
12	O ₂ -Hemoglobin Curve	Bounded	Bounded	Hill	Yes
13	Epidemic SIR (Ebola 2014)	Bounded	Bounded	Hill	Yes
14	Amdahl's Law (CPU Scaling)	Bounded	Bounded	Michaelis-Menten	Yes
15	Muscle Force-Velocity (Hill 1938)	Bounded	Bounded	Exponential	No
16	Facebook MAU Growth	Bounded	Bounded	Logistic	Yes
17	Brownian Diffusion (MSD)	Multiplicative	Power law	Power law	Yes
18	Horton's Law (Stream Numbers)	Additive	Exponential	Exponential	Yes
19	Neural Scaling Laws (LLM Loss)	Multiplicative	Power law	Power law	Yes
20	Time Crystal Order (Rydberg Gas)	Bounded	Bounded	Power law	No
21	Stellar Mass-Luminosity	Multiplicative	Power law	Logistic	No
22	Heart Rate vs Body Mass	Multiplicative	Power law	Power law	Yes
23	Rent's Rule (VLSI Pin Count)	Multiplicative	Power law	Power law	Yes
24	Taylor's Power Law	Multiplicative	Power law	Power law	Yes
25	Hack's Law (Stream Length)	Multiplicative	Power law	Power law	Yes

**PRIMARY RESULT, ON THE SIX-MODEL SET (NO LOGARITHMIC OR PURE LINEAR CANDIDATE;
SECTION 3.2): 19/25 EMPIRICAL DOMAINS CONFIRMED ($p = 7 \times 10^{-5}$, PERMUTATION)**

Under strict AICc-based model selection, the Cauchy-predicted family is confirmed in 19 of 25 empirical curve-fit domains. The marginal-conditioned permutation probability is $p = 7 \times 10^{-5}$; the binomial probability under uniform random assignment to the three families present in this candidate set is $p = 1.56 \times 10^{-5}$, roughly 1 in 64,000; under the four-family grid the same result gives $p = 1.27 \times 10^{-7}$, roughly 1 in 7.9 million, so the figure reported here is the conservative one. On the strictly Cauchy-derived subset alone (multiplicative and additive domains only) the result is 14/18 at permutation $p = 3.0 \times 10^{-4}$: the headline does not depend on the bounded family, whose exhaustiveness remains open.

This is strong enough to publish as exploratory evidence. It is not strong enough to claim as proven. Independent replication with pre-registered operator classification is the necessary next step.

5.1 Baseline-20 rerun

As context, the original 20 baseline domains were rerun with the new proper AICc fitter. The result is 15/20 ($p = 1.67 \times 10^{-4}$). This is not directly comparable to the v1 result of 14/20, because the fitter changed (R^2 -based AIC approximation to proper AICc) and some model selections shifted accordingly. The baseline-20 rerun is reported for transparency, not as evidence that the new fitter is more or less conservative than the old one.

6. The Six Empirical Misses

Six of 25 empirical domains produced a best-fit model outside the predicted family. Each miss has a specific, identifiable explanation. Honest analysis follows.

#	DOMAIN	PREDICTED	AICC BEST	EXPLANATION
1	Kleiber's Law	Power law	Saturation-exp	Classic contested case. Kleiber's original 1932 dataset has only 13 mammals. A bounded model can capture slight upper curvature at extreme body masses. The power-law exponent debate (0.67 vs 0.75) has persisted for decades.
3	Species-Area (Galapagos)	Power law	Hill	Small dataset (30 Galapagos islands) with high scatter. Several islands have very small areas with disproportionate species counts. The Hill function can absorb this scatter with its extra parameter.
6	Zipf's Law	Power law	Michaelis-Menten	Finite-corpus truncation. The Brown Corpus has a hard upper bound on word frequency (69,971 for 'the'). At low ranks, frequencies saturate against this ceiling, pulling AICc towards a bounded model. Zipf's law is a power law in the asymptotic limit; finite corpora violate this limit.

#	DOMAIN	PREDICTED	AICC BEST	EXPLANATION
15	Muscle Force-Velocity	Bounded	Exponential	The Hill 1938 force-velocity data shows a monotonic decrease from maximum isometric force to zero at maximum velocity. The data does not reach or demonstrate an asymptotic plateau - it terminates where force reaches zero. An exponential decay fits this truncated range better than a saturation curve. The bounded prediction requires data spanning the full approach to saturation.
20	Time Crystal Order	Bounded	Power law	Only 4 data points. With so few observations, AICc penalises the extra parameters of saturation models. A power law, with fewer free parameters, wins on parsimony. The bounded prediction may be correct but is untestable with this dataset.
21	Stellar Mass-Luminosity	Power law	Logistic	The mass-luminosity relation shows significant scatter at high stellar masses where radiation pressure, convective instability, and different fusion pathways create deviations from a simple power law. A logistic model absorbs this scatter by fitting an apparent upper rollover.

In summary: three misses involve small or truncated datasets (species-area, time crystal, Zipf); one involves data that does not span the full saturation range (muscle force-velocity); one involves a classically contested relationship (Kleiber); and one involves astrophysical complexity at extreme values (stellar mass-luminosity). None represents a case where the predicted family is fundamentally wrong in principle - but neither can any miss be dismissed. These are genuine failures that an independent replication must address.

7. Secondary Tier: Published Exponents

7.1 Direct published exponents (13/13)

Thirteen taxa with well-established metabolic scaling exponents were tested against the ARC/Cauchy dimensional prediction. The prediction requires three conditions: (a) multiplicative composition (Cauchy's equation constrains the form to a power law), (b) d -dimensional space-filling transport geometry, and (c) a conservation or optimisation constraint on resource flow (energy minimisation in West et al.; supply-demand balance in Banavar et al.; steady-state energy balance in Demetrius). Under all three conditions, the exponent is constrained to $d/(d+1)$. Neither Cauchy alone, nor space-filling alone, is sufficient. Organisms with three-dimensional vascular transport and conserved flow should scale as $M^{3/4}$, those with two-dimensional transport as $M^{2/3}$, and those with one-dimensional filamentous transport as $M^{1/2}$.

NOTE ON THE 3/4 EXPONENT

The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White & Seymour 2003; Glazier 2005, 2022). The largest modern dataset (619 species) gives a maximum-likelihood estimate of 0.687 (95% CI 0.674-0.701), which excludes 0.750. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of the empirical range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4. The framework predicts this variation rather than a single universal exponent.

#	TAXON	PREDICTED	OBSERVED	NEAREST	CI INCLUDES?
26	Mammals	0.750	0.737	$d = 3$	Yes
27	Birds	0.750	0.720	$d = 3$	Yes
28	Fish	0.750	0.800	$d = 3$	Yes
29	Reptiles	0.750	0.760	$d = 3$	Yes
30	Insects	0.750	0.750	$d = 3$	Yes
31	Amphibians	0.750	0.740	$d = 3$	Yes
32	Crustaceans	0.750	0.730	$d = 3$	Yes
33	Jellyfish	0.667	0.680	$d = 2$	Yes
34	Cnidarians	0.667	0.700	$d = 2$	Yes
35	Ctenophores	0.667	0.660	$d = 2$	Yes
36	Ectomycorrhizal fungi	0.500	0.580	$d = 1$	Yes
37	Marine fungi	0.500	0.530	$d = 1$	Yes
38	Saprotrophic fungi (20°C)	0.500	0.530	$d = 1$	Yes

All 13 published exponents fall nearest to the predicted dimensional comparator ($d/(d+1)$), and all 13 confidence intervals include the predicted value. This is strong corroborating evidence, but it tests a different claim (specific exponent) than the primary result (functional family), so it is reported separately.

PROVENANCE NOTE ON THE 13-TAXA TABLE

The table above does not carry a per-row citation column. Section 10.2 references [26]-[35] name the broader compendium literature on interspecific metabolic scaling (West, Brown and Enquist 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2006, 2010; White and Seymour 2003; Glazier 2005, 2022; Kolokotronis et al. 2010) from which the mammalian, vertebrate and invertebrate scaling exponent literature is compiled, but this paper does not currently anchor a locked per-row source citation to each observed exponent or confidence interval in the table. Independent readers should treat individual rows as pending a per-row source audit before reuse in secondary work; the jellyfish, cnidarian and ctenophore rows (33-35) and the three fungal rows (36-38) are the ones for which a per-row source citation has not been anchored in this paper.

CAVEAT ON $d=2$ BIOLOGICAL ENTRIES

The jellyfish, cnidarian, and ctenophore entries (rows 33-35) are classified as nearest-comparator matches to $d = 2$, meaning their observed exponents fall closer to $2/3$ than to either $1/2$ or $3/4$. However, the $d = 2$ biological prediction remains untested in the strong sense because no known organism possesses a genuinely two-dimensional hierarchical space-filling transport network with conserved flow. These organisms lack the vascular architecture that the $d/(d+1)$ derivation assumes. The $d = 2$ confirmation exists in cosmology (Friedmann matter-era solution, exact) and physics (percolation, fragmentation), not in biology. These rows should be understood as empirical proximity to the $d = 2$ comparator, not as dimensional confirmation.

7.2 Provisional published exponents (3/6)

Six additional taxa with contested or proxy-based dimensional assignments were tested. Only 3 of 6 match the nearest predicted comparator. The misses are flatworms (disputed 2D vs 3D transport), bryozoans (colonial organisms with anomalous scaling near 1.0), and glass eels (elongated but retaining 3D vascular transport). These are noted for completeness but carry low evidential weight due to the contested dimensional assignments.

7.3 Analytic identities (6/6)

Six domains defined by exact analytic formulae ($E = mc^2$, hydrogen energy levels, Arrhenius kinetics, Michaelis-Menten kinetics, Friedmann matter-era expansion, de Sitter dark-energy expansion) were included as sanity checks. All 6 confirm. These are tautological - fitting a curve to data generated by an exact formula must recover that formula - and carry no independent evidential weight. They confirm only that the fitter works correctly.

8. Significance Framing

The primary result ($p = 7 \times 10^{-5}$, permutation) deserves honest contextualisation.

- **In absolute terms:** under the uniform-chance binomial null, roughly 1 in 64,000: if 64,000 researchers each picked scaling families at random for 25 domains, on average one would match 19 or more by chance. The stricter permutation null, which concedes that power laws are simply common among real fits, still leaves $p = 7 \times 10^{-5}$.
- **Relative to convention:** both tests land three orders of magnitude beyond $p < 0.05$. Sigma equivalences are not quoted because they shift with the choice of null.
- **What it supports:** the result is strong enough to publish as exploratory evidence that Cauchy-type composition constraints predict scaling families at well-above-chance rates.
- **What it does not support:** it is not strong enough to claim the theory is proven. The particle-physics discovery threshold is 5σ ($p \approx 3 \times 10^{-7}$). The test lacks formal pre-registration. The operator classifications were made by the author, not an independent panel. The candidate set contained no logarithmic and no pure linear model, so the hit rate is conditional on that model set (Section 3.2).
- **What is needed:** independent replication with pre-registered operator classification, ideally by researchers with no stake in the outcome.

9. Group Analysis

9.1 Multiplicative composition: power laws

Of 14 empirical domains classified as multiplicative, 10 produce power-law scaling as their AICc-best model. These span metabolic allometry (heart rate vs body mass), urban economics (Betten-court 2007), technology learning curves (Wright's law), computational linguistics (Heap's law), skill acquisition (Crossman 1959), Brownian motion (Catipovic 2013), neural network scaling (Kaplan 2020), VLSI design (Rent's rule), population ecology (Taylor's power law), and fluvial geomorphology (Hack's law).

Four multiplicative domains miss: Kleiber's law (bounded model edges out power law on a 13-point dataset), species-area (high scatter on small island data), stellar mass-luminosity (scatter at high mass), and Zipf's law (finite-corpus truncation).

9.2 Additive composition: exponentials

All four domains with additive composition operators produce exponential scaling as their AICc-best model: Moore's law (transistor counts), radioactive decay (P-32), the Gutenberg-Richter earthquake frequency law, and Horton's law (stream numbers). This is a clean 4/4.

9.3 Bounded composition: saturation curves

Of seven domains with bounded composition operators, five produce bounded (saturation) models as AICc-best: bacterial growth (logistic), hemoglobin binding (Hill), Ebola epidemic (Hill), Amdahl's law (Michaelis-Menten), and Facebook growth (logistic). Two miss: muscle force-

velocity (exponential decay fits the truncated range better) and time crystal order (only 4 data points).

10. Data Provenance

10.1 Provenance caveat

Three domains use theoretical or reference values rather than independent empirical measurements: radioactive decay (NIST reference half-life for P-32), Amdahl's law (theoretical formula with assumed 10% serial fraction), and Moore's law (Wikipedia-sourced transistor counts from manufacturer specifications). These confirm the framework trivially and should be weighted accordingly.

10.2 Citations for all 25 empirical domains

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11. Related Work

A literature search returned no previous publication that invokes all four Cauchy functional equations as a unifying cross-domain principle constraining the forms of scaling laws. Readers are invited to submit prior work; a genuine antecedent that does this defeats the priority claim of this paper.

The closest existing work includes:

- **Bingham & Ostaszewski (2015):** connect Cauchy's equation to regular variation theory. Pure mathematics. No cross-domain physical science application.
- **West, Brown & Enquist (1997):** derive a 3/4 metabolic exponent from fractal vascular networks with energy-minimised flow. Domain-specific. Does not predict other domains. Note that the empirical exponent is debated (see Warning box in Section 7.1).
- **Banavar, Maritan & Rinaldo (1999, 2010):** derive the same $d/(d + 1)$ form from geometric constraints on transportation networks combined with supply-demand conservation, without

requiring fractal geometry. Independently confirmed the quarter-power exponent from different starting assumptions. Banavar et al. (2010) showed that the exponent ranges from $2/3$ to $3/4$ depending on how velocity scales, with $3/4$ as the optimised maximum - demonstrating that space-filling geometry alone is insufficient without a conservation or optimisation constraint.

- **Demetrius (2003, 2006) and Demetrius & Tuszynski (2010):** derive equivalent scaling from quantum statistical mechanics, applying the Debye model of thermal properties to coupled energy-transducing oscillator networks in d -dimensional space. A completely independent route to the same exponent from different physics (quantum oscillator coupling rather than network geometry). The convergence of this derivation with West's network geometry derivation is explained by Cauchy's multiplicative functional equation applied to multiplicative composition in d -dimensional space with a conservation constraint (see companion paper, On the Origin of Scaling Laws v3.1).
- **Bettencourt (2013):** derives city scaling exponents from network dimensionality, extending the $d/(d+1)$ form beyond biology to urban systems.
- **Zhao (2022):** derives the formula as a universal growth scaling law from network optimisation, providing another independent derivation.
- **He & Chen (2003):** derive metabolic scaling exponents from optimal vascular network geometry with flow conservation constraints.
- **He & Zhang (2004):** extend the derivation to heterogeneous branching networks, recovering the $d/(d+1)$ form under energy-minimisation assumptions.
- **Maino et al. (2014):** derive metabolic scaling from thermodynamic first principles applied to ontogenetic growth, independently obtaining the same exponent family.
- **Geoffrey West, *Scale* (2017):** discusses scaling across biology and cities. No mention of Cauchy's functional equations.
- **Aczél (1966):** standard reference for all four Cauchy equations. Applications to information theory but not to cross-domain scaling.

The gap is genuine. The $d/(d+1)$ formula has been independently derived by at least seven research groups through different mathematical frameworks - fractal networks, geometric constraints with supply-demand balance, quantum metabolism, urban scaling, network optimisation, optimal vascular geometry, and thermodynamic first principles. Every known derivation requires three conditions: multiplicative composition (which Cauchy constrains to the power-law family), d -dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. The mathematical theorems are 200 years old. The biological data is 90 years old. The novel contribution is not the $d/(d+1)$ formula itself (which is well established), nor Cauchy's mathematics, but the claim that Cauchy-type functional equations explain *why* these independent derivations converge on the same form, and that the Cauchy constraint has a physically testable consequence for scaling-law classification across domains.

12. Limitations

- **No pre-registration artefact.** The predictions, data arrays, and fitting logic all reside in a single author-written script. There is no locked pre-registration artefact. This test should be understood as a structured prediction comparison, not a formally registered blind trial.
- **Operator classification by author.** The composition operator classification is done by the author, not by an independent assessor. A replication with independent operator classification would be more rigorous.
- **Bounded composition exhaustiveness is open.** The proof that only saturation-family curves are compatible with bounded composition is not provided and remains an open problem. The bounded category currently groups logistic, Hill, Michaelis-Menten, and hyperbolic forms by empirical convention rather than mathematical proof.
- **Legacy AIC was an approximation.** The v1 test used an R^2 -based AIC approximation ($AIC \approx n \ln(1 - R^2) + 2k$), not a full likelihood-based nonlinear AIC. The v2 test corrects this with proper AICc computed from residual sum of squares.
- **Three datasets use reference values.** Radioactive decay (NIST half-life), Amdahl's law (theoretical formula), and Moore's law (manufacturer specs) use theoretical or reference values rather than independently collected empirical measurements.
- **Family prediction only.** The framework predicts the functional family but not the specific exponent. Predicting exponents requires additional structure (dimensionality, constraint type) developed in the companion Foundational paper.
- **Six empirical misses.** The misses are not dismissible. They identify real limits of the framework: small datasets, truncated ranges, high-scatter regimes, and domains where competing physical effects (radiation pressure, corpus boundaries) perturb the ideal prediction.
- **Operator classification is descriptive, not explanatory.** The framework does not explain *why* a particular domain has a multiplicative rather than additive composition operator. It takes the operator as given and predicts the consequence.
- **Baseline-20 rerun is not cleanly comparable to v1.** The fitter changed from R^2 -based AIC approximation to proper AICc. The model set expanded. A direct comparison of 14/20 (v1) to 15/20 (v2 baseline rerun) is not meaningful because the selection criterion changed.
- **Publication bias risk.** The 5 new expansion domains were selected after the baseline-20 test. While they were chosen to span new physical regimes (stellar astrophysics, allometric heart rate, VLSI, ecology, geomorphology), the selection was not randomised.
- **Finite model set, and specifically two Cauchy families out, one non-Cauchy family in.** The fitter compares six candidate models (power law, exponential, logistic, Hill, Michaelis-Menten, saturation-exponential). Two of the four Cauchy solutions named in Section 2.1 have no candidate (logarithmic $c \log x$; pure linear cx), while the bounded family, whose exhaustiveness remains an open problem, is represented four times. The true best model for any domain may not be in this set, and Section 3.2 states the consequences for the hit counts.
- **Sample size.** Twenty-five empirical domains is a reasonable exploratory sample but not definitive. Expansion to 100+ domains with pre-registered selection criteria would strengthen

or weaken the result.

12.1 Falsifiability: what would defeat this claim

The limitations above describe where the framework is weak; this subsection states the specific outcomes that would falsify it. The central claim, that a domain's composition operator, classified from known physics *before* fitting, predicts its scaling-law family, is defeated by any of the following:

1. **Hit rate at chance.** The primary result is a family-prediction hit rate of 19/25 empirical domains (§5–§6). With three to four candidate families, blind assignment gives a chance baseline of roughly one in three to one in four. The claim is defeated if a replication with independent operator classification returns a hit rate not significantly above that chance baseline.
2. **Post-hoc operator classification.** The test's strength depends on the operator being assignable from independently-known physics without seeing the data. The claim is defeated if, in a blind trial, independent assessors cannot reproduce the author's operator classifications from the physics alone, or if any classification can be shown to have been chosen to fit the observed exponent.
3. **Fit degeneracy.** The prediction is meaningful only if the candidate families are empirically distinguishable on the data at hand. It is defeated for any domain where the fitter reaches the observed curve equally well under the *wrong* family; that is, the family assignment carries no discriminating information for that domain's range and noise.
4. **Unrationalised misses.** Six domains miss (§6). The misses are consistent with the claim only if their causes are fixed as a pre-registered exclusion rule *before* the next test, not supplied after seeing which domains missed. The claim is defeated if the miss rate rises under a pre-registered replication with no post-hoc exclusions.
5. **Four-family re-run.** The drafted four-family confirmatory study (Section 3.2) re-runs this comparison with logarithmic and pure linear candidates included. The claim is weakened if the hit rate falls materially when the missing families are available, and defeated if it falls to the chance baseline; hits that migrate to the logarithmic family will be reported as reclassifications, not as new confirmations.
6. **Exploratory ceiling.** The abstract concedes there is no locked pre-registration artefact. Until one exists (operators classified and predictions locked before fitting, on a randomised domain sample), the claim cannot rise above *exploratory*, and any presentation of it as established would itself falsify the paper's stated evidential status.

What would *not* defeat it: a single additional miss, or a better-fitting model appearing in one domain. The claim is about the aggregate operator-family mapping, not any single fit; but that aggregate must clear the chance baseline under pre-registered, independently-classified replication.

13. Implications

If the framework holds under independent pre-registered replication, it would suggest:

1. The forms of scaling laws are not coincidences but mathematical consequences of composition structure, under the stated axioms.
2. The same principle that governs how a mouse's heart beats governs how neural networks improve with scale.
3. The renormalisation group's power-law predictions and Cauchy's multiplicative equation may be related theorems - a connection that, to the author's knowledge, has not been formally articulated. A formal proof of this equivalence remains open.
4. Any new system's scaling family can be predicted from first principles by classifying its composition operator, without any data fitting.
5. Two hundred years of domain-specific explanations for scaling laws - fractal networks, learning curves, critical phenomena - may be special cases of a single mathematical constraint that Cauchy identified in 1821.

These implications are conditional. The current evidence is exploratory, not definitive.

14. Conclusion

Plant an acorn and given centuries, you get an oak. But plant that oak's acorn, and its acorn, recursively across millennia, and you get a forest that shapes the climate of continents. What we plant in these systems will compound across scales we cannot imagine. *The seed determines the forest.*

Cauchy proved 200 years ago that the seed also determines the *form* of the forest. Multiplicative seeds produce power laws. Additive seeds produce exponentials. Bounded seeds produce saturation. Twenty-five empirical domains tested against this prediction yield 19 confirmations at $p = 7 \times 10^{-5}$ under the marginal-conditioned permutation test, on the six-model candidate set of Section 3.2; the four-family re-run is drafted and awaiting human submission. Thirteen published metabolic exponents fall where the dimensional theory predicts.

The mathematics was always there. We just had not read it as a prediction about the physical world. Whether that reading holds under independent scrutiny is now a question for the scientific community to answer.

Raise AI with care.

15. Reproducibility

The complete validation suite - including all 50 domain definitions, the canonical manifest, and the AICc-based fitting code - is available at:

- Experiment code: github.com/.../experiments/cauchy-unification__Paper-VII
- 50-domain manifest: `canonical_50_domain_manifest.json`

The script requires Python 3.10+, numpy, and scipy. All data is embedded in the manifest with full provenance citations. Output is deterministic.

Corrections and version history

Material scientific changes only. The programme-wide correction record is at </research/corrections/>.

v3.6 (16 August 2026): the four-cell classification replaces the earlier three-family taxonomy throughout, including in the figure captions: a composition operator, a parameter family and a stability statement are distinct objects, and the earlier presentation collapsed them. The withdrawn below-0.2 per cent physics aggregate is removed from live use, and the companion references carry their own identifiers.

v3.4 (14 August 2026): the claim that the logarithmic functional equation governs diminishing returns is struck. Cauchy's classification fixes which functional form a composition rule admits; it does not by itself establish which empirical regime any given system occupies. The classification result is preserved; the empirical inference drawn from it is not.

v3.1 (8 August 2026): the two displayed functional equations in Section 2.2 were corrected: the multiplicative and additive bullets displayed each other's equations. The protocol, the Section 5 table and the validation code were unaffected.

v3.0 (24 March 2026): the space-filling overstatement was corrected by adding the conservation or optimisation constraint as a third necessary condition, and the debate over the mammalian three-quarter exponent is acknowledged rather than assumed away. The domain suite was expanded to fifty across five evidence tiers, the permissive fit criterion was replaced with AICc, and the misses are analysed rather than omitted.

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